

From Text to Traffic: Repurposing Frozen Generative Models for DDoS Detection

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Abstract—Distributed Denial-of-Service (DDoS) attacks remain a critical cybersecurity threat, aiming to disrupt services by overwhelming target networks with volumetric traffic floods. While machine learning models are increasingly deployed for Network Intrusion Detection Systems (NIDS), they traditionally struggle with zero-shot generalization, frequently failing to detect novel, out-of-distribution attack protocols due to rigid global feature scaling. Taking inspiration from the DoLLM framework, we propose an adapted architecture that repurposes a frozen Large Language Model for cross protocol network traffic analysis. Our approach pairs a 1D-Convolutional temporal flow tokenizer and Multihead Attention pooling with a localized sequence-level normalization technique. By normalizing network flows strictly relative to their immediate temporal windows rather than a global dataset baseline, the model achieves scale-invariant reasoning to successfully identify the structural physics of malicious bursts. Evaluated on completely unseen attack protocols, our adapted architecture demonstrates robust cross protocol generalization, achieving a zero-shot F1 score of 0.9922.

Index Terms—Deep Learning, Large Language Models, Distributed Denial-of-Service, Network Security, Network Intrusion Detection Systems

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